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F78AB
Actuarial and Financial Mathematics

Chapter 6: Force of Interest and Continuously-paid cash flows

LEADERS IN IDEAS AND SOLUTIONS

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Introduction to the force of interest

- Consider the case of a nominal compound interest $i^{(m)}$ converted m times a period.
- We can think of the *force of interest*, denoted by δ as the limit of $i^{(m)}$ as $m \rightarrow \infty$, i.e. the nominal interest is compounded infinitely.
- We say that interest is <u>compounded continuously,</u> and have:

$$\delta = \lim_{m \to \infty} i^{(m)}$$

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Force of interest

$$
\begin{aligned}
\delta &= \lim_{m\to\infty} i^{(m)} \\
&= \lim_{m\to\infty} m[(1+i)^{\frac{1}{m}} - 1] \\
&= \lim_{m\to\infty} \frac{\left[(1+i)^{\frac{1}{m}} - 1\right]}{\frac{1}{m}} \\
&= \lim_{m\to 0} \frac{\left[(1+i)^{m} - 1\right]}{m} \\
&= \lim_{m\to 0} \frac{\left[(1+i)^{m}\ln(1+i)\right]}{1} \qquad \text{(Applying L'hopital's rules)} \\
&= \ln(1+i)
\end{aligned}
$$

Or equivalently, $e^{\delta} = 1 + i$

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Continuous accumulation

- If a sum of money $A(0)$ is invested at time 0 and grows continuously, at time t it has grown to $A(t)$
- When the force of interest is constant :

$$A(t) = A(0)\, e^{\delta t} = A(0)\,(1+i)^{t}$$

- The force of interest may not be constant, but instead expressed as a function of time, ie $\delta(t)$, in which case

$$\delta(t) = \frac{1}{A(t)} \frac{dA(t)}{dt}$$

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More generally

- And more generally, for the continuous accumulation between two time periods, t_1 and t_2:

$$A(t_2) = A(t_1) \exp\left(\int_{t_1}^{t_2} \delta(s) ds\right)$$

- We can use this approach to determine both accumulated and present values of cashflows where we are given the force of interest

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Notation

- For times s and t, with $s < t$, denote by:
  - $V(s,t)$ the present value at time s of 1 unit paid at time t
  - $A(s,t)$ the accumulation at time t of 1 unit paid at time s

- For a constant force of interest δ per annum:
  - $V(s,t) = e^{-\delta(t-s)}$.
  - $A(s,t) = e^{\delta(t-s)}$.

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Force of interest Example 1

Exercise 6.1 Given a constant force of interest $\delta(t) = 5\%$ p.a, for all $t \geq 0$, and assuming time is measured in years, calculate

(i) the PV at time $t = 0$ of £150 paid at time $t = 1$.

(ii) the PV at time $t = 0.5$ of £200 paid at time $t = 3.25$.

(iii) the accumulation at time $t = 4$ of £50 paid at time $t = 1$.

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Force of interest example 2

- Given that the annual effective rate of interest is 12%, calculate d, $d^{(4)}$, $d^{(12)}$, $i^{(4)}$, $i^{(12)}$ and δ

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Force of interest example 3

a) Find the accumulated amount at time 20 of a cashflow of £1000 at time 0, assuming at force of interest $\delta(t)$ *p.a.* given by

$$\delta(t) = 0.06 + 0.0005t - 0.00005t^2$$

b) Find the accumulated amount at time 20 of a cashflow of £1000 at time 0, and a further cashflow of £1000 at time 10, assuming at force of interest $\delta(t)$ *p.a.* given by

$$\delta(t) = 0.06 + 0.0005t - 0.00005t^2$$

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Force of interest example 4

Calculate the present value at time 0 of a cashflow of 100 at time 4, and 1000 at time 10, assuming a force of interest per time unit given as follows:

$$\delta = \begin{cases} 0.04 & 0 \leq t < 5 \\ 0.14 - 0.02t & 5 \leq t \end{cases}$$

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Continuously paid cashflows

- Consider an annuity of £1 per year, paid continuously during the year
- We might interpret this as the limit of an annuity paid m times each year, ie

$$a_{\overline{n}|}^{(m)} = \frac{1-(1+i)^{-n}}{i^{(m)}}, \text{ when } m \to \infty$$

- New notation:

$$\bar{a}_{\overline{n}|} = a_{\overline{n}|}^{(\infty)} = \frac{1-(1+i)^{-n}}{i^{(\infty)}} = \frac{1-(1+i)^{-n}}{\delta}.$$

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Continuously paid cashflows

- We also have:

$$\overline{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{\delta} \qquad \text{and} \qquad \overline{a}_{\overline{\infty}|} = \frac{1}{\delta}\,.$$

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Continuously paid cashflows

- Unlike discrete payments, here we are interested in the **rate** at which the payments are being made.

- The rate may be constant, eg £100 pa: or
- may vary over time, ie. be paid at continuously at some rate $\rho(t)$ p.a.

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General case

- Consider a cashflow paid continuously at rate $\rho(t)$ p.a. for T years starting at time 0.
- Partition the time interval [0,T) into timing segments, each of length dt
- Consider one of the segments [t, t+dt] – if dt is very small then the total payments between time t and t+dt is:

$$M(t+dt)_ - M(t) \approx \rho(t)\, dt$$

Where M(t) denotes the total payment between time 0 and time t

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General case, continued

- We treat the amount paid between *t* and *t+dt* as paid exactly at time *t*,
- *Ie we treat the continuous payment made over [t, t+dt)* as a discrete cashflow, $\rho(t)\,dt$, paid at time t.
- PV at time t=0 of the discrete cashflow is V(0,t) $\rho(t)\,dt$
- Repeat for all the tiny intervals, them sum them to find the total PV at time 0.
- Letting $dt \to 0$, the present value at time 0 of the entire cashflow between time 0 and T is:

$$\int_0^T \rho(t)V(0,t)dt = \int_0^T \rho(t)\exp\left(-\int_0^t \delta(u)du\right)dt.$$

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Continuously paid cashflows cont..

Example 1 Calculate the PV at time 0 of a cashflow paid continuously at the rate of £1,000 p.a. from time 0 to time 10 years. Assume that the force of interest is 5% p.a

Example 2 Calculate the PV at time 0 of a cashflow paid continuously from time 0 to time 2 years at the rate $\rho(t) = e^{-0.05t}$ p.a. for all t in the time interval [0,2] using force of interest is 3% p.a.

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Payment over any time period

- If we have cashflows paid continuously at rate $\rho(t)$ p.a. starting for T years, starting at time S, partition the time interval [S,T) into timing segments, each of length dt
- Consider one of the segments [t, t+dt) – if dt is very small then the total payments between time t and t+dt is:

$$M(t+dt)_ - M(t) \approx \rho(t)\, dt$$

- We treat the amount paid between t and $t+dt$ as paid exactly at time t
- The PV at time r of the discrete cashflow is $V(r,t)\ \rho(t)\, dt$

- Repeat for all the tiny intervals, them sum them to find the total PV at time r.
- Letting $dt \rightarrow 0$, the present value at time r of the entire cashflow between time S and T is:

$$\int_S^T \rho(t)V(r,t)dt = \int_S^T \rho(t)\exp\left(-\int_r^t \delta(u)du\right)dt.$$

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Example 3 Calculate the PV at time 2 of a cashflow paid continuously at the rate of £1,000 p.a. from time 5 to time 20 years. assuming a constant force of interest $\delta = 3\%$ p.a.

Example 4 Find the present value at time t=1 of a cashflow paid continuously at the constant rate of £10,000 pa from time 3 to time 14, assuming a constant force of interest $\delta = 6\%$ p.a.

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Accumulated values

- Use the same approach - consider cashflows paid continuously at rate $\rho(t)$ p.a. starting for T years, starting at time S, partition the time interval [S,T) into timing segments, each of length dt
- Consider one of the segments [t, t+dt) – if dt is very small then the total payment between time t and t+dt is approximately $\rho(t)\,dt$

- The AV at time r of the discrete cashflow is $A(t,T)\,\rho(t)\,dt$

- Repeat for all the tiny intervals, them sum them to find the total PV at time T.
- Letting $dt \rightarrow 0$, the accumulated value at time T of the entire cashflow between time S and T is:

$$\int_S^T \rho(t)A(t,T)dt = \int_S^T \rho(t)\exp\left(\int_t^T \delta(u)du\right)dt$$

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Accumulated values examples

Example 1 Calculate the AV at time t=14 of a cashflow paid continuously at the rate of £10,000 p.a. from time 3 to time 14 years. Assume that the constant force of interest is 3% p.a.

Example 2 Calculate the AV at time 12 of a cashflow paid continuously at the rate of £100 p.a. from time 3 to time 5 years. Assume that the force of interest is 4% p.a.

Example 3 Calculate the PV at time 2 of a cashflow paid continuously from time 0 to time 5 years at a rate of £50 p.a., given the force of interest is:

$$\delta(t) = \begin{cases} 0.04 & \text{for } 0 \leq t < 5 \\ 0.03 & \text{for } t \geq 5. \end{cases}$$

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Increasing cashflows

Example

- A cashflow is paid continuously from time 0 to 10 years. The initial rate of payment of £1000 p.a., and increases continuously at the rate of 3%p.a. effective. Calculate the PV at time 0 of the cashflow, assuming a force of interest of 5% p.a.

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Increasing cashflows

Generalising

The PV at the beginning of a cashflow which is paid continuously for *n* years and has a rate of payment *k* units p.a. in year *k* is $(I\bar{a})_{\overline{n}|}$

Where

$$
\begin{aligned}
(I\bar{a})_{\overline{n}|} &= \bar{a}_{\overline{1}|} + 2v\bar{a}_{\overline{1}|} + 3v^2\bar{a}_{\overline{1}|} + \cdots nv^{n-1}\bar{a}_{\overline{1}|} \\
&= \bar{a}_{\overline{1}|}\left(1 + 2v + 3v^2 + \cdots + nv^{n-1}\right) \\
&= \bar{a}_{\overline{1}|}(I\ddot{a})_{\overline{n}|} \\
&= \frac{1-v}{\delta}\frac{\ddot{a}_{\overline{n}|} - nv^n}{d} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{\delta}, \qquad \text{since } d = 1 - v.
\end{aligned}
$$

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Continuously increasing cashflows

Generalising

Consider a cashflow which is paid continuously for *n* years, with rate of payment $\rho(t) = t$, *where* $t \in [0,n]$

The PV of this cashflow is denoted by $(\bar{I}\bar{a})_{\overline{n}|}$

And $$(\bar{I}\bar{a})_{\overline{n}|} = \frac{\bar{a}_{\overline{n}|} - nv^n}{\delta}.$$

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Important formulae

$$\bar{s}_{\overline{n}|} = (1+i)^n \bar{a}_{\overline{n}|}$$

$$(I\bar{s})_{\overline{n}|} = (1+i)^n (I\bar{a})_{\overline{n}|}$$

$$(\bar{I}\bar{s})_{\overline{n}|} = (1+i)^n (\bar{I}\bar{a})_{\overline{n}|}.$$

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Example

Find the value at time 15 of a cashflow paying continuously at a rate $\rho(t)$ p.a. where:

$$\rho(t) = \begin{cases} 10 & \text{for } 0 \leq t < 5 \\ 12 & \text{for } 5 \leq t < 15 \end{cases}$$

Assuming a constant force of interest δ

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Final Example

An investor buys an annuity which pays cash continuously for 5 years. In the first year he receives payment of £3,000 p.a., and thereafter annual payments increase by £500 per annum.

Calculate the purchase price of the annuity, assuming that the annual effective rate of interest is 6% p.a.